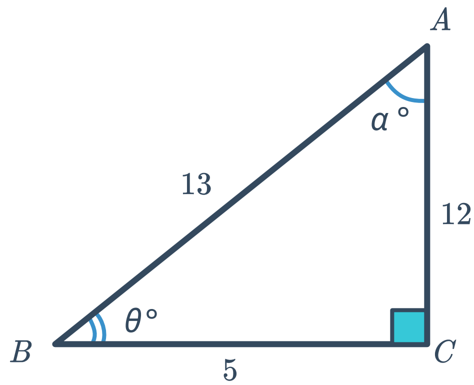
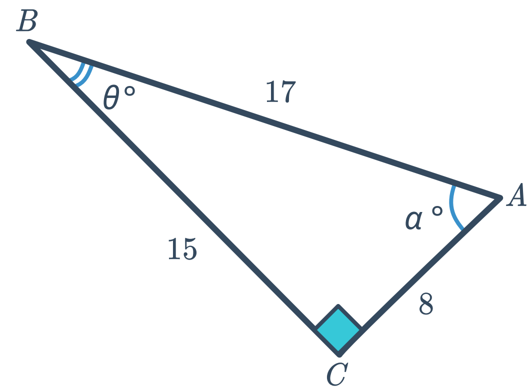
Trigonometric ratios

1 Evaluate $\cos\theta$ for each of the following triangles:

a

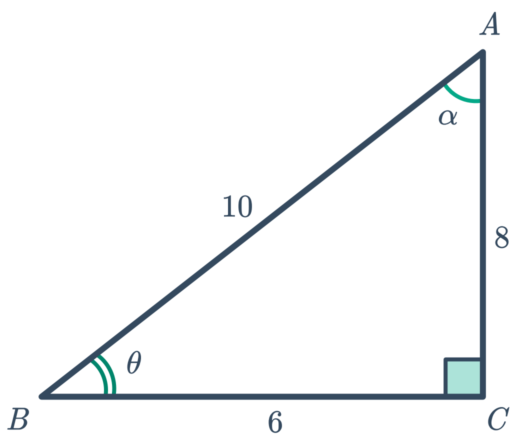


b

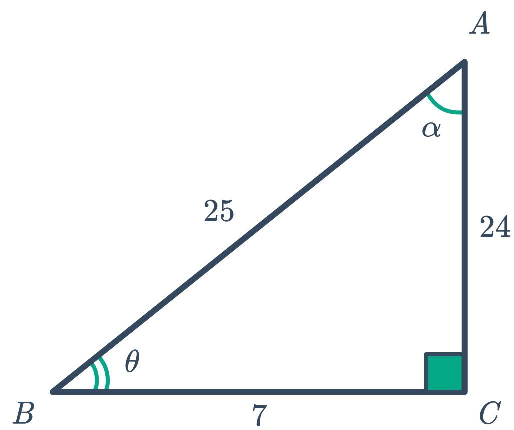


2 Evaluate $\sin\theta$ for each of the following triangles:

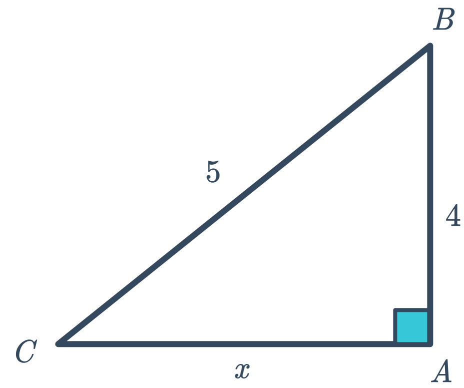
a



b



3 In the following triangle, $\sin\theta = \dfrac{4}{5}$:

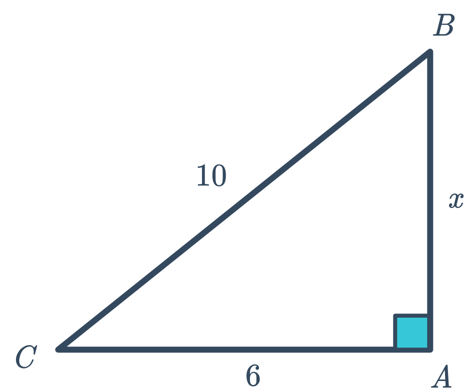


a Which angle is represented by θ?

b Find $\cos\theta$.

c Find $\tan\theta$.

4 In the following triangle, $\cos\theta = \dfrac{6}{10}$:

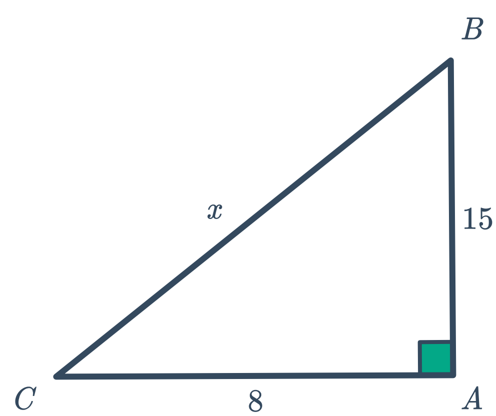


a Which angle is represented by θ?

b Find $\sin\theta$.

c Find $\tan\theta$.

5 In the following triangle, $\tan\theta = \dfrac{15}{8}$:



a Which angle is represented by θ?

b Find $\cos\theta$.

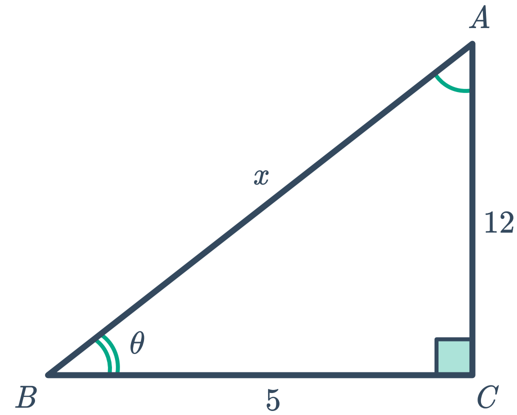
c Find $\sin\theta$.

6 Consider the following triangle:

a Find the value of x.

b Now, find $\sin\theta$.

c Now, find $\cos\theta$.

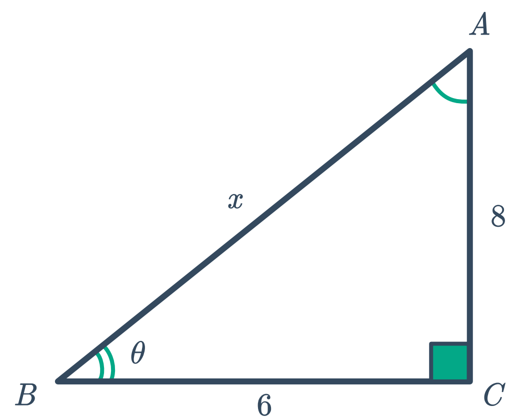


7 Consider the following triangle:

a Find the value of x.

b Find $\sin\theta$.

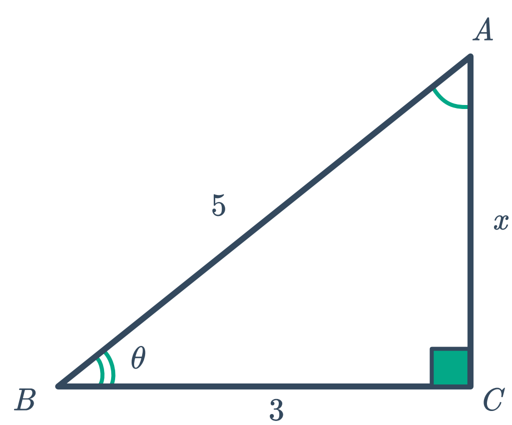
c Find $\cos\theta$.



8 Consider the following triangle:

a Find the value of x.

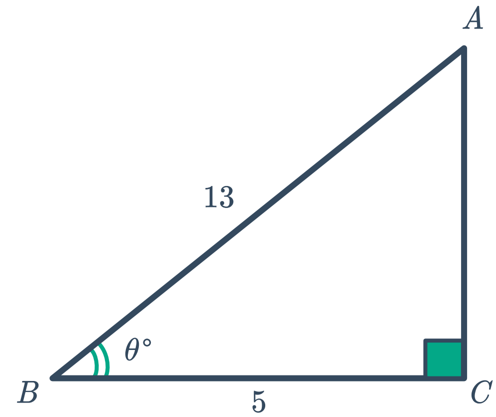
b Now, find $\tan\theta$.



9 Consider the following triangle:

a Find the length of AC.

b Find $\tan\theta$.

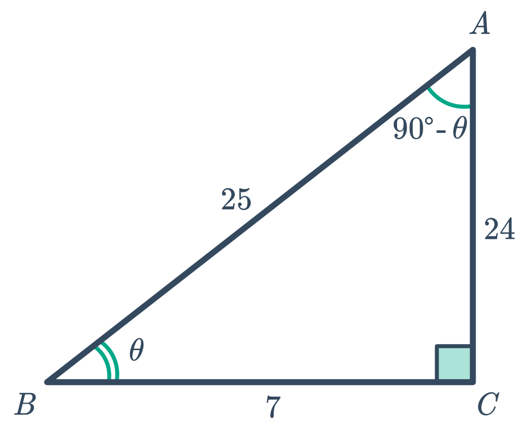


10 Consider the following triangle:

a Find $\cos\left(90^\circ - \theta\right)$ as a fraction.

b Find $\sin\theta$ as a fraction.

c What do you notice?

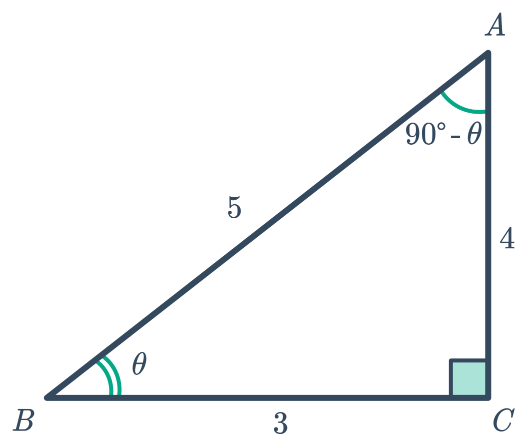


11 Consider the following triangle:

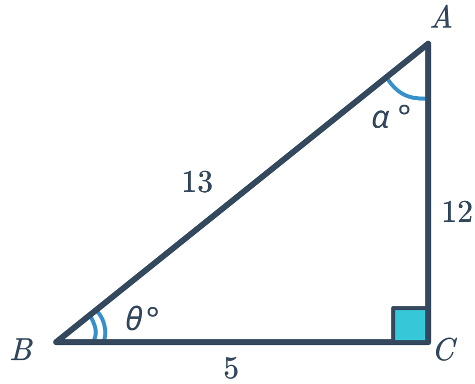
a Find $\sin\left(90^\circ - \theta\right)$ as a fraction.

b Find $\cos\theta$ as a fraction.

c What do you notice?



12 Consider the following triangle:



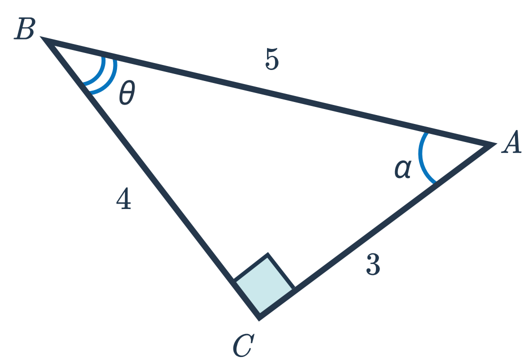
a. Evaluate $\dfrac{\sin\theta}{\cos\theta}$.

b. Evaluate $\tan\theta$.

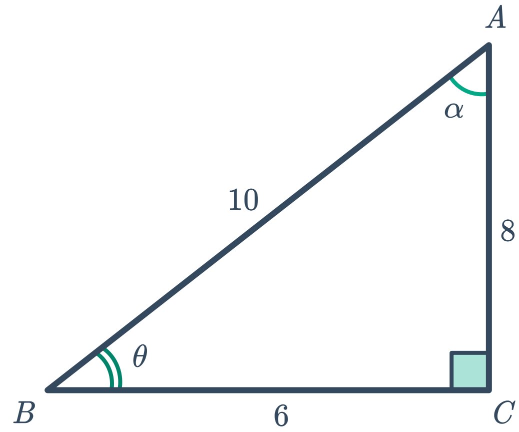
c. What do you notice?

Reciprocal trigonometric ratios

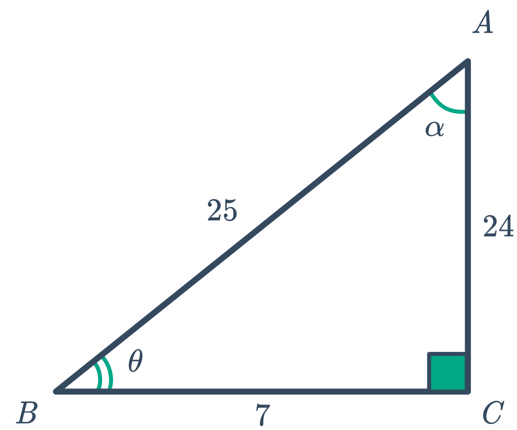
13 Find $\sec\theta$ for the following triangle:



14 Find $\csc\theta$ for the following triangle:

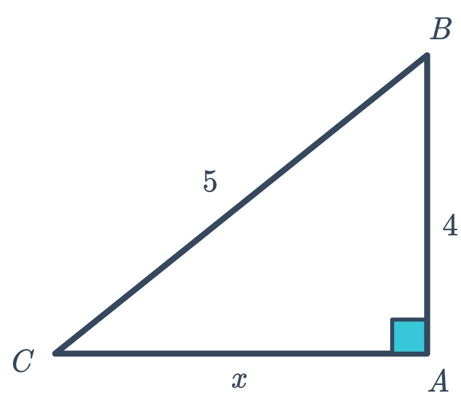


15 Find $\cot\theta$ for the following triangle:



16 In the following triangle, $\csc\theta = \dfrac{5}{4}$:

- **a** Which angle is represented by θ?
- **b** Find the value of x.
- **c** Find $\sec\theta$.
- **d** Find $\cot\theta$.



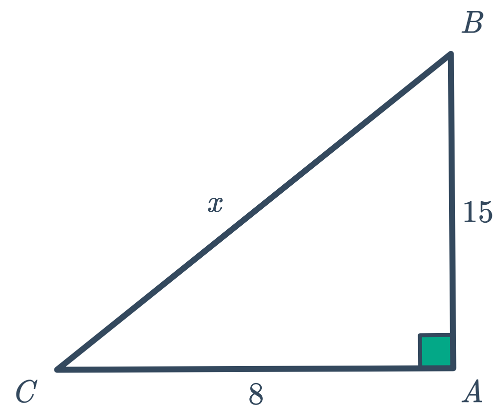
17 In the following triangle, $\cot\theta = \frac{8}{15}$:

a Which angle is represented by θ?

b Find the value of x.

c Find $\sec\theta$.

d Find $\csc\theta$.

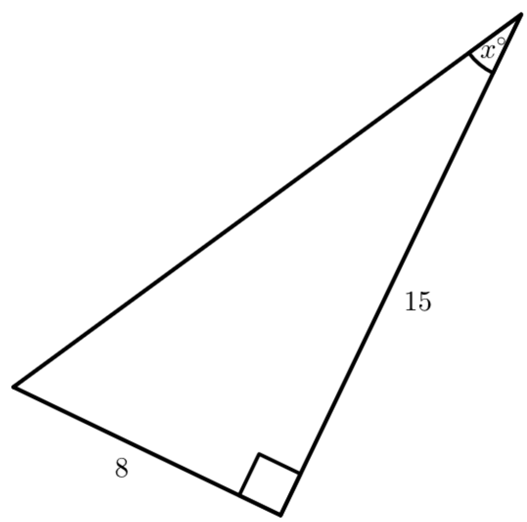


18 Consider the following triangle:

a Calculate the length of the missing side.

b Find the following trigonometric ratios:

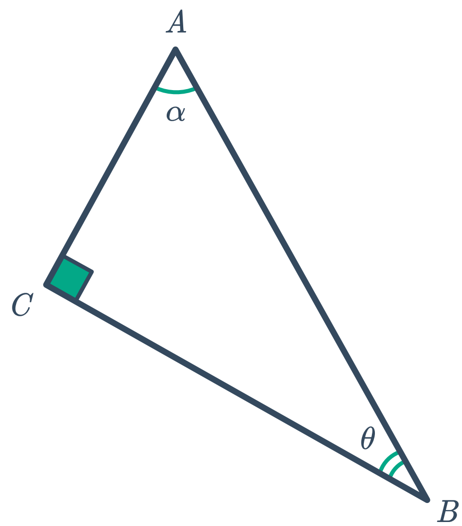
i	$\sin x$	**ii**	$\cos x$
iii	$\tan x$	**iv**	$\sec x$
v	$\csc x$	**vi**	$\cot x$



19 Find the indicated ratios for the following tr[redacted]es:

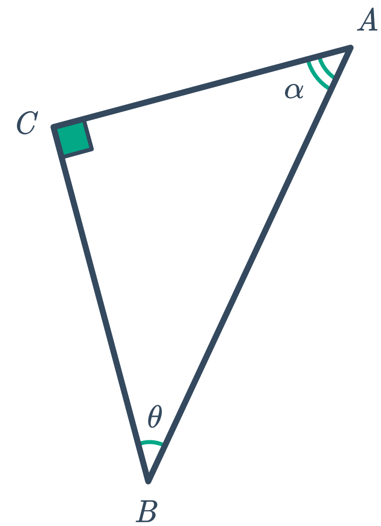
a

i $\tan\theta$ ii $\sin\alpha$



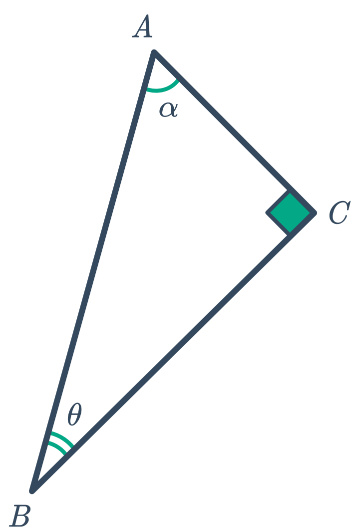
b

i $\sin\theta$ ii $\tan\alpha$



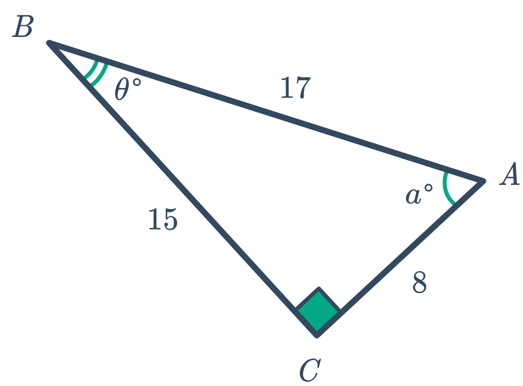
c

i $\cos\theta$ ii $\cos\alpha$



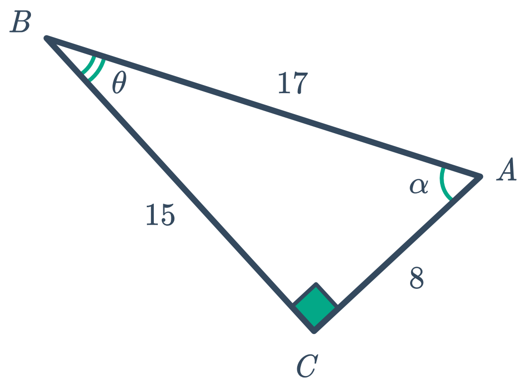
d

i $\tan\theta$ ii $\sin\alpha$

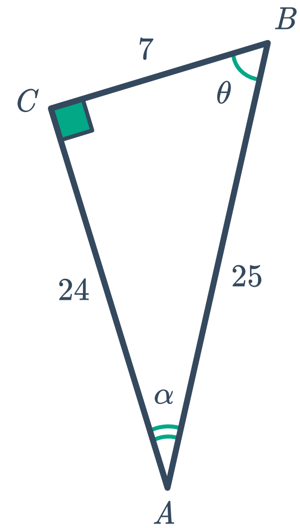


e

i $\cos\theta$ **ii** $\tan\alpha$



i $\sin\theta$ **ii** $\cos\alpha$



20 Consider the following triangle:

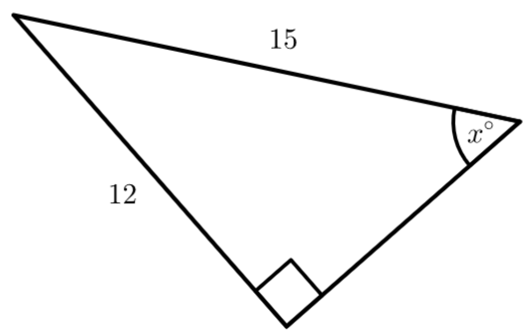
a Calculate the length of the missing side.

b Find the following trigonometric ratios:

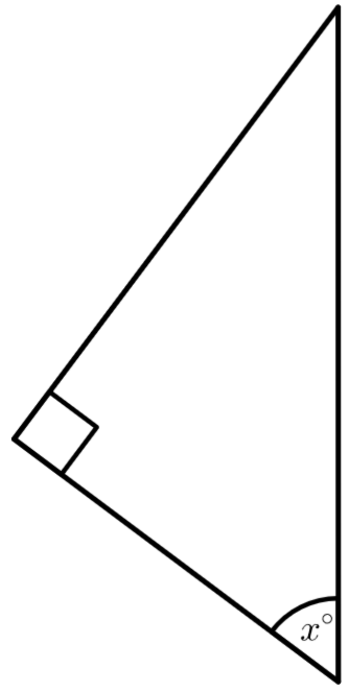
i $\sin x$ **ii** $\cos x$

iii $\tan x$ **iv** $\sec x$

v $\csc x$ **vi** $\cot x$



21 Consider the following triangle:



a Given that $\tan x = \frac{3}{4}$, find $\cos x$.

b Given that $\sin x = \frac{15}{17}$, find $\tan x$.

c Given that $\cos x = \frac{12}{13}$, find $\sin x$.

d Given that $\sin x = \frac{4}{5}$, find $\csc x$.

e Given that $\tan x = \frac{7}{4}$, find $\cot x$.

f Given that $\cos x = \frac{5}{13}$, find $\sec x$.